

PAGE-1 HCI Focus: Interaction & Usability Elements

Human-Computer Interaction (HCI) examines how users navigate, control, and communicate with software interfaces. In Coohom, the interface prioritizes visual affordances, structured layout controls, and reduced cognitive load to simplify 3D spatial editing.

1. Asset Search & Catalog Filter Bar

Allows users to quickly search and filter 3D models, reducing cognitive load while searching for furniture.



3. 2D / 3D Viewport Mode Switcher

Gives a clear visual affordance to switch seamlessly between 2D layout planning and 3D spatial inspection.

2. Canvas Zoom & Navigation Controls

Provides direct manipulation tools to adjust zoom percentages and frame views for precise spatial editing.

What is one simple HCI change you would make to improve the user interface?

One simple change would be to add **hover tooltips with short keyboard shortcuts** to the top toolbar icons (Save, Undo, Redo, Clear, Toolset, Gallery). New users often hesitate because the icons are small and unlabeled, and tooltips would teach them the function without leaving the screen. A second small improvement would be a **collapsible "Beginner Mode"** that hides advanced panels like the multi-floor editor and material settings until the user is ready. Both changes cost little to implement but noticeably improve discoverability and confidence for first-time users.

PAGE-2 CG Focus: Rendering & 3D Visual Elements

Computer Graphics (CG) focuses on geometric modeling, digital lighting shaders, and scene rendering techniques. In Coohom, graphics algorithms compute 3D mesh boundaries, light behavior, and surface textures to simulate realistic spatial depth.

2. Directional Lighting & Soft Shadows

Simulates a virtual sunlight source to cast soft realistic shadows, giving essential visual depth cues.

3. Texture & Material Mapping

Maps 2D bitmap images onto 3D polygon surfaces to give real-world leaf patterns, glass reflection, and surface realism.



1. 3D Mesh Bounding Box

Displays the 3D spatial boundary and oriented vector geometry used for object selection and collision bounds.

How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

High-quality CG in Coohom — realistic sunlight, soft shadows, textured materials, and perspective depth — helps users instantly understand spatial relationships, so they can judge room size, furniture placement, and lighting without reading measurements. For example, the shadows under the roof and hedges in Screenshot 2 make the 3D form readable at a glance, which speeds up design decisions. However, heavy graphics can also hinder interaction: on lower-end devices, real-time rendering may lag, and the dense visual detail can sometimes compete with UI elements, making small buttons harder to spot. In short, good graphics support interaction when they clarify space, but hinder it when they slow performance or clutter the interface.